

NETWORK TECHNOLOGY

UNIT- 5 Mail Services

5.1 Application Layer services:

5.1.1 concepts of email

5.1.2 working of email account and services

5.1.3 URL and URL types (Absolute, Relative)

5.2 Case study of email:

5.2.1 From sender to receiver (Mailer, Mail Server, Mailbox)

5.2.2 Functionality and use of protocols at different layers

5.3 Case study of locating Website:

5.3.1 URL and locating URL

5.3.2 Steps and protocols involved in accessing URL

5.3.3 Concepts of search engine and purpose.

5.1 Application Layer services:

★ Concepts of email

Definition: - Email is a service which allows us to send the message in electronic mode over the internet. It offers an efficient, inexpensive and real time means of distributing information among people.

❖ E-Mail Address

- Each user of email is assigned a unique name for his email account. This name is known as an E-mail address. Different users can send and receive messages according to the email address.
- E-mail is generally of the form username@domainname.
- For example,

- **For Example :-** webmaster@tutorialspoint.com is an e-mail address where webmaster is username and tutorialspoint.com is domain name.

The username and the domain name are separated by @ (at) symbol.

- E-mail addresses are not case sensitive.
- Spaces are not allowed in email addresses.

Components of an Email:

- ❖ **Sender:** The sender creates an email in which he records the information that needs to be transferred to the receiver.
- ❖ **Receiver:** The receiver gets the information sent by the sender via email.
- ❖ **Email address:** An email address is just like a house address where the communication arrives for the sender and receiver and they communicate with each other.
- ❖ **Mailer:** The mailer program allows the ability to read, write, manage and delete the emails like Gmail, Outlook, etc.
- ❖ **Mail Server:** The mail server is responsible for sending, receiving, managing, and recording all the data processed by their respective mail programs and then processing them to their respective users.
- ❖ **SMTP:** SMTP stands for [Simple mail transfer protocol](#). SMTP basically uses the internet network connection to send and receive email messages over the Internet.

❖ E-mail System

➤ E-mail system comprises of the following three components:

- Mailer
- Mail Server
- Mailbox

→ Mailer

It is also called mail program, mail application or mail client. It allows us to manage, read and compose email.

→ Mail Server

The function of the mail server is to receive, store and deliver the email. It is necessary for mail servers to be Running all the time because if it crashes or is down, email can be lost.

→ Mailboxes

Mailbox is generally a folder that contains emails and information about them.

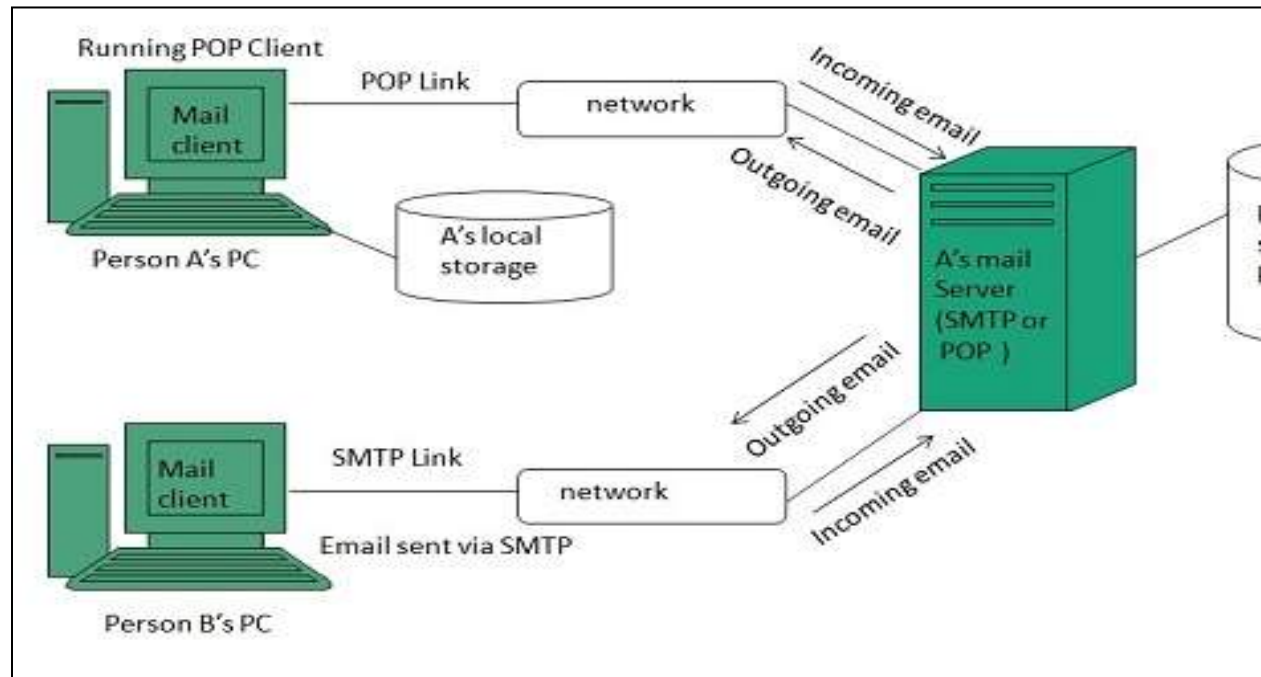
❖ Working of E-mail

- Email working follows the client server approach. In this client is the mailer i.e. the mail application or mail program and server is a device that manages emails.

- Following example will take you through the basic steps involved in sending and receiving emails and will give you a better understanding of working of email system:
 - Suppose person A wants to send an email message to person B.
 - Person A composes the messages using a mailer program i.e. mail client and then select Send option.
 - The message is routed to Simple Mail Transfer Protocol to person B's mail server.
 - The mail server stores the email message on disk in an area designated for person B.

- Now, suppose person B is running a POP client and knows how to communicate with B's mail server.
- It will periodically poll the POP server to check if any new email has arrived for B. As in this case, person B has sent an email for person B, so email is forwarded over the network to B's PC. This message is now

stored on person B's PC.



❖ URL and its URL types (Absolute, Relative)

- Every document on the Web has a unique address. This address is known as Uniform Resource Locator (URL).

→ URL Elements

A URL is made up of several parts, each of which offers information to the web browser to help find the page.

It is easier to learn the parts of a URL, if you look at the example URL given below, there are three key parts: the [scheme](#), the [host address](#), and the [file path](#). The following section will discuss each of them:

type://address/path

**Basic Structure
of URL**

➤ The Scheme

The scheme identifies the type of protocol and URL you are linking to and therefore, how the resource should be retrieved. For example, most web browsers use Hypertext Transfer Protocol (HTTP) to pass information to communicate with the web servers and this is the reason a URL starts with `http://`.

Example:- `ftp://`, `http://`, `https://`, `file://`

Sr.No	Scheme & Description
1	<code>http://</code> Hypertext Transfer Protocol (HTTP) is used to request pages from Web servers and send them back from Web servers to browsers.

2	https:// Secure Hypertext Transfer Protocol (HTTPS) encrypts the data sent between the browser and the Web server using a digital certificate.
3	ftp:// File Transfer Protocol is another method for transferring files on the Web. While HTTP is a lot more popular for viewing Web sites because of its integration with browsers, FTP is still commonly used protocol to transfer large files across the Web and to upload source files to your Web server.
4	file:// Used to indicate that a file is on the local hard disk or a shared director

➤ **The Host Address**

The host address is where a website can be found, either the IP address (four sets of numbers between 0 and 255, for example 68.178.157.132) or more commonly the domain name for a site such as www.tutorialspoint.com. Note that "www" is not actually part of the domain name although it is often used in the host address.

➤ **The File Path**

The filepath always begins with a forward slash character, and may consist of one or more directory or folder names. Each directory name is separated

by forward slash characters and the file path may end with a filename at the end. Here index.htm is the filename which is available in html directory:

<https://www.tutorialspoint.com/html/index.htm>

❖ **Types of URL:** URL gives the address of files created for web pages or other documents like an image, pdf for a doc file, etc.

→ There are two types of URL:

- Absolute URL
- Relative URL

❖ **Absolute** – An absolute URL is the complete address of a resource.

- This type of URL contains both the domain name and directory/page path. An absolute URL gives complete location information.
- It begins with a protocol like “http://” and continues, including every detail. An absolute URL typically comes with the following syntax.

For example http://www.tutorialspoint.com/html/html_text_links.htm

❖ **Relative** –

- A relative URL indicates where the resource is in relation to the current page. Given URL is added with the <base> element to form a complete URL.

- This type of URL contains the path excluding the domain name. Relative means “in relation to”, and a relative URL tells a URL location in terms of the current location.
- Relative path is used for reference to a given link of a file that exists within the same domain.

For example - /html/html_text_links.htm

❖ Case study of email

★ From Sender to Receiver:

- The sender first needs the email address of the receiver to send the information to be communicated via email. When the sender writes all the information in the email along with the email address of the receiver and clicks on the send button, the mail program transfers the message to the MTA (Mail Transfer Agent) which is transferred from the local computer of the sender to the mail server via the SMTP protocol.
- Then the webmail server looks out for the similar mail transfer agent of the receiver and locates it whether it is using the same DNS (domain name server) or a different service. The DNS looks for the mail exchanger service of the receiver.
- Now, the SMTP protocol transfers the message between both mail servers through their mailing agents. Then the receiver's MTA finally transfers this message to the receiver's local computer.
- In case, the receiver uses POP protocol then when he receives the email, then the copy of the email at the webserver will get deleted. And if he uses IMAP then the copy of the email gets stored on the webserver and it can be changed at any time by the user.

❖ Protocols of Email:

- Emails basically use two types of standard protocols for communication over the Internet. They are:-
 - POP
 - IMAP

1. POP:

- a. POP stands for [post office protocol](#) for email. Similar to a post office, our approach is just to drop the email over the service mail provider and then leave it for services to handle the transfer of messages.
- b. We can even be disconnected from the Internet after sending the email via POP. Also, there is no requirement of leaving a copy of the email over the web server as it uses very little memory.
- c. POP allows concentrating all the emails from different email addresses to accumulate on a single mail program. Although, there are some disadvantages of POP protocol like the communication medium is unidirectional, i.e it will transfer information from sender to receiver but not vice versa.

2. IMAP:

- a. IMAP stands for [Internet message access protocol](#). IMAP has some special advantages over POP like it supports bidirectional communication over email and there is no need to store

conversations on servers as they are already well-maintained in a database.

- b. It has some advanced features like it tells the sender that the receiver has read the email sent by him.**